

Fall 25, 4100 Problem Set 7, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Let $A \in \mathbb{C}^{n \times n}$ and W be a subspace of $\mathbb{C}^n$. Recall that W is an A -invariant subspace if and only if for all $\mathbf{x} \in W$, $A\mathbf{x} \in W$. Prove that the generalized eigenspace, $\text{null}(A - \lambda I)^n$ is A -invariant for any eigenvalue λ and n .

let $x \in \text{null}(A - \lambda I)^n$

by definition $(A - \lambda I)^n x = 0$

$A = (A - \lambda I) + \lambda I$

$Ax = ((A - \lambda I) + \lambda I)x$

because both are polynomials in A they commute $(A - \lambda I)A = A(A - \lambda I)$

$(A - \lambda I)^n A = A(A - \lambda I)^n$

$(A - \lambda I)^n (Ax) = A(A - \lambda I)^n x$

but $(A - \lambda I)^n x = 0$

so $A(A - \lambda I)^n x = A \cdot 0 = 0$

and $(A - \lambda I)^n (Ax) = 0$

therefore $Ax \in \text{null}(A - \lambda I)^n$

QED

2. For each of the following matrices find the eigenvalues, eigenvectors, and generalized eigenvectors, and state the Jordan form

a. $\begin{bmatrix} -3 & -1 \\ 1 & -5 \end{bmatrix}$

$$\det(A - \lambda I) = \det \begin{bmatrix} -3 - \lambda & -1 \\ 1 & -5 - \lambda \end{bmatrix} = (-3 - \lambda)(-5 - \lambda) + 1 = \lambda^2 + 8\lambda + 16 = (\lambda + 4)^2$$

$\lambda = -4$ with algebraic multiplicity 2

$$(A + 4I)x = 0$$

$$A + 4I = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

one dimensional null space $x_1 = x_2$

$$\text{eigenvector } v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

geometric multiplicity = 1

$$(A + 4I)v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$J = \begin{bmatrix} -4 & 1 \\ 0 & -4 \end{bmatrix}$$

b. $\begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -5 \\ 0 & 0 & 3 \end{bmatrix}$

$$\det(B - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 1 & 3 \\ -1 & 0 - \lambda & -5 \\ 0 & 0 & 3 - \lambda \end{bmatrix} = (2 - \lambda)(-\lambda)(3 - \lambda) - 0 + 0 - (1)(-1)(3 - \lambda) + 0 - 0 = (2 - \lambda)(-\lambda)(3 - \lambda) + (3 - \lambda) = -\lambda^3 + 5\lambda^2 - 7\lambda + 3 = (\lambda - 1)^2(\lambda - 3)$$

$\lambda = 1, 3$ with algebraic multiplicities 2 and 1 respectively

$$(B - I)x = 0$$

$$B - I = \begin{bmatrix} 1 & 1 & 3 \\ -1 & -1 & -5 \\ 0 & 0 & 2 \end{bmatrix}$$

$$(B - I)u_1 = 0$$

$$u_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$(B - 3I)x = 0$$

$$B - 3I = \begin{bmatrix} -2 & 1 & 3 \\ -1 & -3 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(B - 3I)u_2 = 0$$

$$u_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$(B-I)w = u_1$$

$$w = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$J = \text{diag}(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, [\ 3 \])$$

$$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

3. a. Given $J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$, find a formula for the matrix exponential e^{Jt} .

let $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and write $J = \lambda I + N$

note $N^2 = 0$ and λI commutes with N

$$e^{Jt} = e^{(\lambda I + N)t} = e^{\lambda t} e^{Nt}$$

$$e^{Nt} = I + Nt$$

$$e^{Jt} = e^{\lambda t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

b. Use the result above and 2a. to solve the linear system

$$\begin{aligned} x_1' &= -3x_1 - x_2 \\ x_2' &= x_1 - 5x_2 \end{aligned}$$

with initial values $x_1(0) = 1, x_2(0) = 2$

let $A = \begin{bmatrix} -3 & -1 \\ 1 & -5 \end{bmatrix}$ and write $x' = Ax$

A is similar to $J = \begin{bmatrix} -4 & 1 \\ 0 & -4 \end{bmatrix}$ with similarity matrix $P = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $P^{-1} =$

$$\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A = PJP^{-1} \text{ so } e^{At} = Pe^{Jt}P^{-1}$$

$$e^{Jt} = e^{-4t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$e^{At} = P(e^{-4t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix})P^{-1} = e^{-4t} P \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} P^{-1}$$

$$e^{At} = e^{-4t} \begin{bmatrix} 1+t & -t \\ t & 1-t \end{bmatrix}$$

$$x(t) = e^{At}x(0) = e^{-4t} \begin{bmatrix} 1+t & -t \\ t & 1-t \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = e^{-4t} \begin{bmatrix} 1+t-2t \\ t+2(1-t) \end{bmatrix} = e^{-4t} \begin{bmatrix} 1-t \\ 2-t \end{bmatrix}$$

$$x_1(t) = (1-t)e^{-4t}$$

$$x_2(t) = (2-t)e^{-4t}$$

4. Let A be such that $p_A(\lambda) = |A - \lambda I| = (\lambda - 4)(\lambda - 2)^5(\lambda + 1)^2$ with $\text{nullity}(A - 4I) = 1$, $\text{nullity}(A - 2I) = 3$, and $\text{nullity}(A + I) = 1$. Find all possible Jordan forms of A up to reordering.

$$\lambda = 4 \text{ a mult} = 1$$

$$\lambda = 2 \text{ a mult} = 5$$

$$\lambda = -1 \text{ a mult} = 2$$

$$\text{nullity}(A - 4I) = 1 \text{ one Jordan block for } \lambda = 4$$

$$J_1(4)$$

$$\text{nullity}(A - 2I) = 3 \text{ one Jordan block for } \lambda = 2, \text{ can be } 3 + 1 + 1 \text{ or } 2 + 2 + 1$$

$$J_3(2) \oplus J_1(2) \oplus J_1(2) \text{ or } J_2(2) \oplus J_2(2) \oplus J_1(2)$$

$$\text{nullity}(A + I) = 1 \text{ one Jordan block for } \lambda = -1$$

$$J_2(-1)$$

two possible Jordan forms (up to reordering of blocks)

$$J = J_1(4) \oplus (J_3(2) \oplus J_1(2) \oplus J_1(2)) \oplus J_2(-1)$$

$$J = J_1(4) \oplus (J_2(2) \oplus J_2(2) \oplus J_1(2)) \oplus J_2(-1)$$

5. Let A be such that $p_A(\lambda) = |A - \lambda I| = (\lambda - 3)^7(\lambda + 2)^6$ with minimal polynomial, $\mu_A(x) = (x - 3)^5(x + 2)^3$ and geometric multiplicities, $\text{nullity}(A - 3I) = 2$ and $\text{nullity}(A + 2I) = 3$. Find the **unique** (up to reordering) Jordan form of A .

$\lambda = 7$, a mult 7, g mult 2, largest block size 5, must be $5 + 2$

$J_5(3)$ and $J_2(3)$

$\lambda = -2$, a mult 6, g mult 3, largest block size 3, must be $3 + 2 + 1$

$J_3(-2)$, $J_2(-2)$, and $J_1(-2)$

$J = \text{diag}(J_5(3), J_2(3), J_3(-2), J_2(-2), J_1(-2))$

6. Let $A \in \mathbb{R}^{n \times n}$ be such that $A^k = I$. Prove that A is diagonalizable. (Hint: a polynomial p has a simple zero at z_0 iff $p(z_0) = 0$ but $p'(z_0) \neq 0$.)

set $p(x) = x^k - 1$

then $p(A) = A^k - I = 0$

so the minimal polynomial $\mu_A(x)$ of A divides $p(x)$

the roots of $p(x)$ are the k th roots of unity ζ

for each root we have $p'(\zeta) = k\zeta^{k-1}$

$\zeta \neq 0$ and $k \neq 0$ so $p'(\zeta) \neq 0$

by the hint, every root of p is a simple root, no repeated factors

therefore square free, so any divisor including $\mu_A(x)$ is also square free

therefore the minimal polynomial has one simple distinct linear factors

therefore A is diagonalizable

QED